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# Chapter 6

## Quadrature

### 6.1 Preliminaries

**Numerical quadrature** (or numerical integration) constitutes a broad family of algorithms for calculating the numerical value of a definite integral. The basic problem considered by numerical integration is to compute an approximate solution to a definite integral:

$$\int_a^b f(x) dx.$$

**Definition 1** *A quadrature rule is an approximation formula of the type*

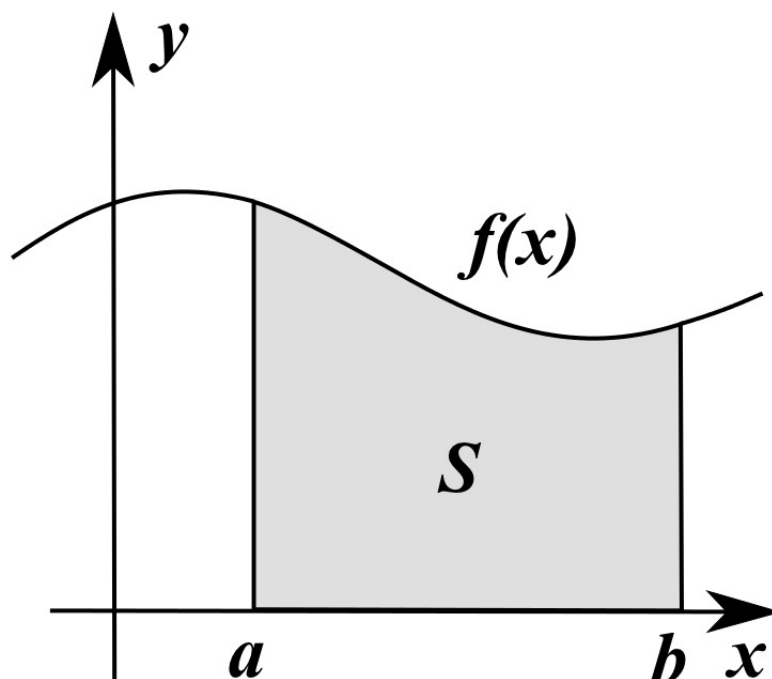
$$\int_a^b f(x) dx \approx \sum_{j=1}^m w_j f(x_j) = w_1 f(x_1) + w_2 f(x_2) + \cdots + w_m f(x_m). \quad (6.1.1)$$

*We will always assume that the points satisfy*

$$a \leq x_1 < x_2 < x_3 < \cdots < x_m \leq b.$$

*The coefficients  $w_j$  are called the weights.*

There are several reasons for carrying out numerical integration. The integrand  $f(x)$  may be known only at certain points, such as obtained by sampling. A formula for the integrand may be known, but it may be difficult or impossible to find an antiderivative which is an elementary function. An example of such an integrand is  $f(x) = \exp(-x^2)$ , the antiderivative of which cannot be written in elementary form.



## 6.2 Newton-Cotes rules or interpolatory rules

### 6.2.1 Simple rules

We can generate a quadrature rule by choosing the points in some way and then replacing the integrand by (the Lagrange form of) its interpolating polynomial, i.e., by using the approximation

$$f(x) \approx \sum_{j=1}^m f(x_j) \ell_j(x) \quad \text{where} \quad \ell_j(x) = \prod_{\substack{k=1 \\ k \neq j}}^m \frac{x - x_k}{x_j - x_k}.$$

In this way, we have

$$\int_a^b f(x) dx \approx \int_a^b \sum_{j=1}^m f(x_j) \ell_j(x) dx = \sum_{j=1}^m w_j f(x_j)$$

where

$$w_j = \int_a^b \ell_j(x) dx \quad \text{for } j = 1, 2, \dots, m.$$

**Midpoint rule:** If we choose

$$m = 1 \quad \text{and} \quad x_1 = \frac{a+b}{2} = \text{midpoint of } [a, b],$$

then  $\ell_1(x) = 1$  so the weight is  $w_1 = b - a$ , giving the *midpoint rule*

$$\int_a^b f(x) dx \approx (b - a) f\left(\frac{a + b}{2}\right).$$

**Trapezoidal rule:** If we choose

$$m = 2, \quad x_1 = a, \quad x_2 = b,$$

then

$$\ell_1(x) = \frac{x - b}{a - b} \quad \text{and} \quad \ell_2(x) = \frac{x - a}{b - a},$$

so the weights are

$$\begin{aligned} w_1 &= \int_a^b \frac{x - b}{a - b} dx = \left[ \frac{(x - b)^2}{2(a - b)} \right]_a^b = -\frac{(a - b)^2}{2(a - b)} = \frac{b - a}{2}, \\ w_2 &= \int_a^b \frac{x - a}{b - a} dx = \left[ \frac{(x - a)^2}{2(b - a)} \right]_a^b = \frac{(b - a)^2}{2(b - a)} = \frac{b - a}{2}, \end{aligned}$$

giving the *trapezoidal rule*

$$\int_a^b f(x) dx \approx \frac{b - a}{2} [f(a) + f(b)],$$

so called because the right-hand side is the area of the trapezoid with corners  $(a, 0)$ ,  $(b, 0)$ ,  $(b, f(b))$  and  $(a, f(a))$ .

**Simpson's rule:**

**Example 6.2.1** Show that choosing

$$m = 3, \quad x_1 = a, \quad x_2 = \frac{a + b}{2}, \quad x_3 = b,$$

yields *Simpson's rule*:

$$\int_a^b f(x) dx \approx \frac{b - a}{6} \left[ f(a) + 4f\left(\frac{a + b}{2}\right) + f(b) \right].$$

With  $x_1 = a$ ,  $x_2 = (a + b)/2$  and  $x_3 = b$ , the Lagrangians are

$$\begin{aligned} \ell_1(x) &= \frac{(2x - a - b)(x - b)}{(a - b)^2} \\ \ell_2(x) &= \frac{-4(x - a)(x - b)}{(a - b)^2} \\ \ell_3(x) &= \frac{(2x - a - b)(x - a)}{(a - b)^2} \end{aligned}$$

Hence

$$\begin{aligned} w_1 &= \int_a^b \ell_1(x) dx = -\frac{1}{6} \frac{(a-b)^3}{(a-b)^2} = \frac{1}{6}(b-a) \\ w_2 &= \int_a^b \ell_2(x) dx = -\frac{2}{3} \frac{(a-b)^3}{(a-b)^2} = \frac{2}{3}(b-a) \\ w_3 &= \int_a^b \ell_3(x) dx = -\frac{1}{6} \frac{(a-b)^3}{(a-b)^2} = \frac{1}{6}(b-a) \end{aligned}$$

### Relation between the midpoint, trapezoidal, and Simpson rules:

Notice that if we write

$$\begin{aligned} M(f) &= (b-a)f\left(\frac{a+b}{2}\right), & T(f) &= \frac{b-a}{2}[f(a) + f(b)], \\ S(f) &= \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right], \end{aligned}$$

then

$$S(f) = \frac{2}{3}M(f) + \frac{1}{3}T(f),$$

i.e., Simpson's rule is a weighted average of the midpoint and trapezoidal rules.

### 6.2.2 Degree of precision

**Definition 2** Let us denote the error in the approximation (6.1.1) by

$$E(f) = \int_a^b f(x) dx - \sum_{j=1}^m w_j f(x_j),$$

so that

$$\int_a^b f(x) dx = \sum_{j=1}^m w_j f(x_j) + E(f).$$

We say that the quadrature rule has degree of precision  $r$  if

$$E(1) = E(x) = E(x^2) = \cdots = E(x^r) = 0 \quad \text{but} \quad E(x^{r+1}) \neq 0.$$

Since  $E(f)$  is linear in  $f$ , this means that the degree of precision is  $r$  iff the rule is exact for every polynomial of degree  $r$  or less, but not for polynomials of degree  $r+1$ .

**Example 6.2.2** Show that the midpoint and trapezoidal rules both have degree of precision 1.

With the midpoint rule, we have

$$E(1) = \int_a^b 1dx - (b-a) = (b-a) - (b-a) = 0,$$

and

$$E(x) = \int_a^b xdx - (b-a)\frac{(a+b)}{2} = \frac{1}{2}(b^2 - a^2) - (b-a)\frac{(a+b)}{2} = 0,$$

while

$$E(x^2) = \int_a^b x^2dx - (b-a)\frac{(a+b)^2}{2^2} = \frac{1}{3}(b^3 - a^3) - (b-a)\frac{(a+b)^2}{2^2} \neq 0.$$

With the trapezoidal rule, we have

$$E(1) = \int_a^b 1dx - \frac{(b-a)}{2}(1+1) = b-a - (b-a) = 0,$$

and

$$E(x) = \int_a^b xdx - \frac{(b-a)}{2}(a+b) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(b^2 - a^2) = 0,$$

while

$$E(x^2) = \int_a^b x^2dx - \frac{(b-a)}{2}(a^2 + b^2) = \frac{1}{3}(b^3 - a^3) - \frac{(b-a)}{2}(a^2 + b^2) \neq 0.$$

**Example 6.2.3** Show that Simpson's rule has degree of precision 3.

For Simpson's rule

$$E(1) = \int_a^b 1dx - \frac{b-a}{6}(1+4+1) = b-a - (b-a) = 0,$$

and

$$E(x) = \int_a^b xdx - \frac{b-a}{6}(a+2(a+b)+b) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(b^2 - a^2) = 0,$$

and

$$\begin{aligned} E(x^2) &= \int_a^b x^2dx - \frac{b-a}{6}(a^2 + 4((a+b)^2/4) + b^2) \\ &= \frac{1}{3}(b^3 - a^3) - \frac{(b-a)}{3}(a^2 + ab + b^2) = 0, \end{aligned}$$

and

$$\begin{aligned} E(x^3) &= \int_a^b x^3dx - \frac{b-a}{6}(a^3 + 4((a+b)^3/8) + b^3) \\ &= \frac{1}{4}(b^4 - a^4) - \frac{(b-a)}{6} \frac{3}{2}(a^3 + a^2b + ab^2 + b^3) = 0. \end{aligned}$$

but

$$E(x^4) \neq 0.$$

**Theorem 6.2.1** *The degree of precision of an  $m$ -point interpolatory quadrature rule is at least  $m - 1$ . In case  $m$  is odd and the points are symmetric with respect to the midpoint of the interval  $[a, b]$ , the degree of precision is at least  $m$ .*

**Proof** Let  $f(x) = x^k$  then

$$I(f) = I(x^k) = \int_a^b f(x)dx = \int_a^b x^k dx$$

An  $m$ -point Newton-Cotes rules applied to  $f(x) = x^k$  has the form

$$Q(f) = Q(x^k) = \sum_{j=1}^m w_j f(x_j) = \sum_{j=1}^m w_j x_j^k,$$

where

$$w_j = \int_a^b \ell_j(x)dx, \text{ for } j = 1, \dots, m,$$

where  $\ell_j(x)$  are the Lagrange polynomials defined on  $x_1, \dots, x_m$ . Hence

$$\begin{aligned} E(x^k) &= I(x^k) - Q(x^k) = \int_a^b x^k dx - \sum_{j=1}^m w_j x_j^k \\ &= \int_a^b x^k dx - \sum_{j=1}^m \left( \int_a^b \ell_j(x)dx \right) x_j^k \\ &= \int_a^b \left( x^k - \sum_{j=1}^m \ell_j(x) x_j^k \right) dx. \end{aligned}$$

For  $k \leq m - 1$  the Lagrangian interpolation polynomials of  $x^k$  are identical to  $x^k$ , hence

$$E(x^k) = \int_a^b 0 = 0, \text{ for } k \leq m - 1.$$

For  $k = m$ , using Theorem 4.3.1 in Chapter 4 with  $f(x) = x^m$ , (hence  $m = n + 1$  where  $n + 1$  points was used in Theorem 4.3.1),  $f^{(m)}(x) = m!$ , we have

$$x^m - \sum_{j=1}^m \ell_j x_j^m = \frac{m!}{m!} (x - x_1)(x - x_2) \cdots (x - x_m).$$

Hence

$$E(x^m) = \int_a^b \left( x^m - \sum_{j=1}^m \ell_j(x) x_j^m \right) dx = \int_a^b (x - x_1) \cdots (x - x_m) dx := \int_a^b \phi(x) dx,$$

with  $\phi(x) = (x - x_1) \cdots (x - x_m)$ .

If the points  $x_1, \dots, x_m$  are symmetric w.r.t the midpoint  $(a+b)/2$  then by define the following transform

$$\tau = x - (a+b)/2, \quad \Phi(\tau) := \phi(x),$$

we can show that  $\Phi(\tau) = -\Phi(-\tau)$ . For example, take  $m = 3$ ,  $x_1 = a$ ,  $x_2 = a + h$ ,  $x_3 = a + 2h = b$ . Then with  $\tau = x - (a+b)/2 = x - x_2$  we have

$$\begin{aligned} \Phi(\tau) &:= \phi(x) = (x - x_1)(x - x_2)(x - x_3) \\ &= (\tau + x_2 - x_1)\tau(\tau + x_2 - x_3) \\ &= (\tau + h)\tau(\tau - h) = \tau(\tau^2 - h^2). \end{aligned}$$

On the other hand,  $\Phi(-\tau) = (-\tau + h)(-\tau)(-\tau - h) = \tau(h^2 - \tau^2) = -\Phi(\tau)$ . So  $\Phi(\tau)$  is an odd function. Therefore

$$\int_a^b \phi(x) dx = \int_{(a-b)/2}^{(b-a)/2} \Phi(\tau) d\tau = 0,$$

since the integral of an odd continuous function over a symmetric region always equals zero.  $\square$

### 6.2.3 Quadrature error

In the following, for any function  $f$  defined on  $[a, b]$  we write

$$I(f) = \int_a^b f(x) dx, \quad Q(f) = \sum_{j=1}^m w_j f(x_j) \quad \text{and} \quad E(f) = I(f) - Q(f).$$

**Definition 3** The Peano kernel for the rule  $Q(f)$  of precision  $r$  is defined to be

$$K(t) = E(\phi_t), \quad t \in [a, b],$$

where for any  $t \in [a, b]$  the function  $\phi_t$  is defined by

$$\phi_t(x) = \frac{(x-t)_+^r}{r!} = \begin{cases} \frac{(x-t)^r}{r!} & \text{if } x \geq t, \\ 0 & \text{if } x < t. \end{cases}$$

#### Error via Peano kernel

**Theorem 6.2.2** If the quadrature rule  $Q$  has degree of precision  $r \geq 1$ , and if the integrand  $f$  is  $C^{r+1}$  on the interval  $[a, b]$ , then the error may be written as

$$E(f) = \int_a^b K(t) f^{(r+1)}(t) dt.$$

Hence, if  $K(t)$  does not change sign on the interval  $[a, b]$ , there exists  $c \in [a, b]$  such that

$$E(f) = M f^{(r+1)}(c),$$

where  $M := \int_a^b K(t) dt$ .

**Proof.** Since the rule has degree of precision  $r$ , we use the Taylor expansion of  $f$  to write  $f = P_r + R$ , where  $P_r$  is a polynomial of degree  $r$  and the remainder  $R(x)$  is given by

$$R(x) = \int_a^b \frac{(x-t)_+^r}{r!} f^{(r+1)}(t) dt,$$

where

$$(x-t)_+^r = \begin{cases} (x-t)^r & \text{if } a \leq t \leq x, \\ 0 & \text{if } x < t \leq b. \end{cases}$$

Using linearity and the fact that the quadrature has order of precision  $r$ , we have

$$\begin{aligned} E(f) &= I(f) - Q(f) = I(P_r) + I(R) - Q(P_r) - Q(R) \\ &= I(P_r) - Q(P_r) + I(R) - Q(R) \\ &= E(P_r) + E(R) = E(R). \end{aligned}$$

We now compute each term in  $E(R)$  separately

$$\begin{aligned} I(R) &= \int_a^b \left( \int_a^b \frac{(x-t)_+^r}{r!} f^{(r+1)}(t) dt \right) dx \\ &= \int_a^b \left( \int_a^b \frac{(x-t)_+^r}{r!} dx \right) f^{(r+1)}(t) dt \\ &= \int_a^b \left( \int_t^b \frac{(x-t)^r}{r!} dx \right) f^{(r+1)}(t) dt. \end{aligned}$$

On the other hand

$$\begin{aligned} Q(R) &= \sum_{j=1}^m w_j \int_a^b \frac{(x_j-t)_+^r}{r!} f^{(r+1)}(t) dt \\ &= \int_a^b \left( \sum_{j=1}^m w_j \frac{(x_j-t)_+^r}{r!} \right) f^{(r+1)}(t) dt \\ &= \int_a^b \left( \sum_{x_j > t}^m w_j \frac{(x_j-t)^r}{r!} \right) f^{(r+1)}(t) dt. \end{aligned}$$

It follows that

$$\begin{aligned} E(R) &= \int_a^b \left( \int_t^b \frac{(x-t)^r}{r!} dx - \sum_{x_j > t}^m w_j \frac{(x_j-t)^r}{r!} \right) f^{(r+1)}(t) dt \\ &= \int_a^b K(t) f^{(r+1)}(t) dt, \end{aligned}$$

proving the theorem. □

**Example 6.2.4** Show that the Peano kernel for a rule of precision  $r$  satisfies

$$K(t) = \int_t^b \frac{(x-t)^r}{r!} dx - \sum_{x_j > t} w_j \frac{(x_j-t)^r}{r!}.$$

**Example 6.2.5** Show that the Peano kernel for the trapezoidal rule is

$$K(t) = -\frac{1}{2}(t-a)(b-t).$$

For trapezoidal rule

$$\begin{aligned} K(t) &= \int_t^b (x-t) dx - \frac{(b-a)}{2}(b-t) \\ &= \frac{1}{2}(b-t)^2 - \frac{(b-a)}{2}(b-t) = -\frac{1}{2}(b-t)(t-a). \end{aligned}$$

**Example 6.2.6** Show that the Peano kernel for the midpoint rule is

$$K(t) = \begin{cases} \frac{1}{2}(t-a)^2 & \text{for } a \leq t < \frac{1}{2}(a+b), \\ \frac{1}{2}(b-t)^2 & \text{for } \frac{1}{2}(a+b) \leq t \leq b. \end{cases}$$

If  $a \leq t < \frac{1}{2}(a+b)$  then

$$\begin{aligned} K(t) &= \int_t^b (x-t) dx - (b-a)((b+a)/2 - t) \\ &= \frac{1}{2}(t^2 + a^2 - 2at) = \frac{1}{2}(t-a)^2. \end{aligned}$$

If  $(a+b)/2 \leq t \leq b$  then

$$K(t) = \int_t^b (x-t) dx = \frac{1}{2}(b-t)^2.$$

**Example 6.2.7** Show that for the midpoint rule,

$$E(f) = \frac{(b-a)^3}{24} f''(c).$$

For the midpoint rule

$$\begin{aligned} \int_a^b K(t)dt &= \frac{1}{2} \int_a^{(a+b)/2} (t-a)^2 dt + \frac{1}{2} \int_{(a+b)/2}^b (b-t)^2 dt \\ &= \frac{1}{6} (t-a)^3 \Big|_{t=a}^{t=(a+b)/2} - \frac{1}{6} (b-t)^3 \Big|_{t=(a+b)/2}^{t=b} \\ &= \frac{1}{48} (b-a)^3 + \frac{1}{48} (b-a)^3 = \frac{1}{24} (b-a)^3. \end{aligned}$$

Therefore

$$E(f) = \frac{(b-a)^3}{24} f''(c).$$

**Example 6.2.8** Show that for the trapezoidal rule,

$$E(f) = -\frac{(b-a)^3}{12} f''(c).$$

For the trapezoidal rule,

$$\begin{aligned} \int_a^b K(t)dt &= -\frac{1}{2} \int_a^b (t-a)(b-t) dt \\ &= \frac{1}{12} (a-b)^3. \end{aligned}$$

Therefore,

$$E(f) = -\frac{(b-a)^3}{12} f''(c).$$

**Example 6.2.9** Show that for Simpson's rule,

$$K(t) = \begin{cases} \frac{1}{72} (t-a)^3 [3(t-a) - 2(b-a)] & \text{for } a \leq t < \frac{1}{2}(a+b), \\ \frac{1}{72} (b-t)^3 [3(b-t) - 2(b-a)] & \text{for } \frac{1}{2}(a+b) \leq t \leq b, \end{cases}$$

and

$$E(f) = -\frac{(b-a)^5}{2880} f^{(4)}(c).$$

For the Simpson's rule, for  $a \leq t < \frac{(a+b)}{2}$ , we have

$$\begin{aligned} K(t) &= \int_t^b \frac{(x-t)^3}{3!} dx - \frac{2(b-a)}{3} \frac{((a+b)/2 - t)^3}{3!} - \frac{(b-a)}{6} \frac{(b-t)^3}{3!} \\ &= \frac{1}{72} (t-a)^3 [3(t-a) - 2(b-a)] \end{aligned}$$

For  $\frac{(a+b)}{2} \leq t < b$ , we have

$$\begin{aligned} K(t) &= \int_t^b \frac{(x-t)^3}{3!} dx - \frac{(b-a)}{6} \frac{(b-t)^3}{3!} \\ &= \frac{1}{72} (b-t)^3 [3(b-t) - 2(b-a)]. \end{aligned}$$

For  $t = b$ , it is easy to see that

$$K(b) = \int_b^b \frac{(x-t)^3}{3!} dx = 0.$$

For the error, we compute

$$\begin{aligned} \int_a^b K(t) dt &= \frac{1}{72} \int_a^{(a+b)/2} (t-a)^3 [3(t-a) - 2(b-a)] dt + \\ &\quad \frac{1}{72} \int_{(a+b)/2}^b (b-t)^3 [3(b-t) - 2(b-a)] dt \\ &= -\frac{(b-a)^5}{2880}. \end{aligned}$$

So,

$$E(f) = -\frac{(b-a)^5}{2880} f^{(4)}(c).$$

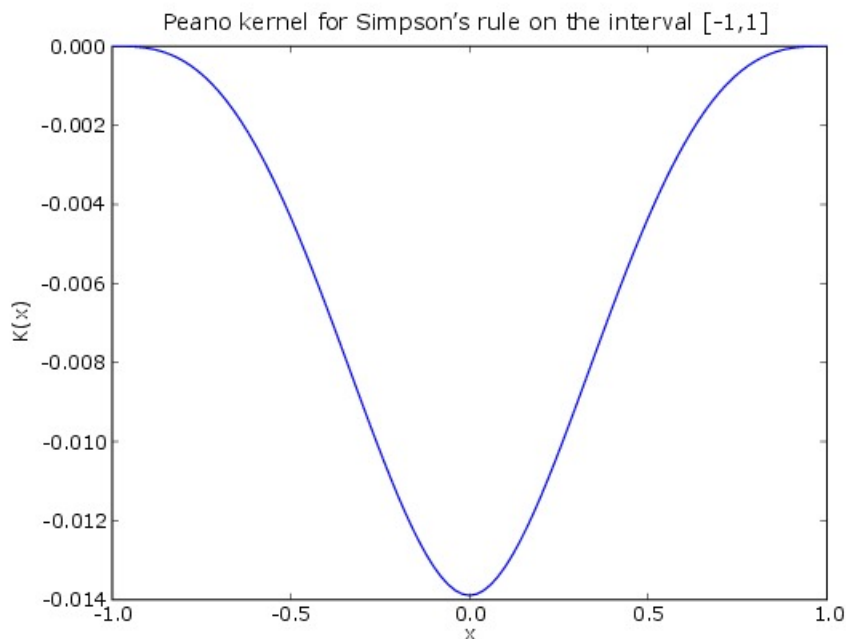
Note that

$$a \leq t \leq \frac{a+b}{2} \iff \frac{a+b}{2} \leq a+b-t \leq b,$$

and that

$$K(t) = K(a+b-t) \quad \forall t \in [a, b].$$

This means the graph of  $K$  is symmetric about  $(a+b)/2$ .



### 6.2.4 Composite Rules

**Composite Rules** To achieve higher accuracy, we can subdivide the interval  $[a, b]$  and apply a basic quadrature rule over each subinterval.

Suppose

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b$$

and for  $j = 1, 2, \dots, n$  put

$$h_j = x_j - x_{j-1} \quad \text{and} \quad x_{j-1/2} = \frac{x_j + x_{j-1}}{2}.$$

**Composite midpoint rule** Applying the midpoint rule over the subinterval  $[x_{j-1}, x_j]$  gives

$$\int_{x_{j-1}}^{x_j} f(x) dx \approx h_j f(x_{j-1/2}),$$

and summing over  $j$  gives the *composite midpoint rule*

$$\int_a^b f(x) dx = \sum_{j=1}^n \int_{x_{j-1}}^{x_j} f(x) dx \approx \sum_{j=1}^n h_j f(x_{j-1/2}).$$

**Composite midpoint rule with uniform subdivision** If we use a uniform subdivision

$$x_j = a + (j-1)h \quad \text{for } j = 0, 1, 2, \dots, n, \quad \text{where } h = \frac{b-a}{n},$$

so that the *step-size*  $h_j = h$  is the same for all  $j$ , then the composite midpoint rule can be written as

$$\int_a^b f(x) dx \approx h \sum_{j=1}^n f(x_{j-1/2}). \quad (6.2.1)$$

The total error is just the sum of the errors for each subinterval:

$$E(f) = \int_a^b f(x) dx - \sum_{j=1}^n h_j f(x_{j-1/2}) = \sum_{j=1}^n E_j(f)$$

where

$$E_j(f) = \int_{x_{j-1}}^{x_j} f(x) dx - h_j f(x_{j-1/2}).$$

With  $h_j = h$  for  $j = 1, \dots, n$ , we have

$$\sum_{j=1}^n E_j(f) = \frac{h^3}{24} f''(c_j), \quad c_j \in (x_{j-1}, x_j).$$

Using intermediate value theorem, we can show that there is a  $c \in [a, b]$  so that

$$\frac{f''(c_1) + \dots + f''(c_n)}{n} = f''(c).$$

Hence, using  $nh = (b - a)$ , we obtain

$$\sum_{j=1}^n E_j(f) = n f''(c) \frac{h^3}{24} = \frac{(b - a)}{24} f''(c) h^2.$$

In summary, we have the following theorem.

**Theorem 6.2.3** *Consider the composite midpoint rule (6.2.1) with uniform step-size  $h$ . If  $f$  is  $C^2$  on the interval  $[a, b]$ , then there is a number  $c \in [a, b]$  such that*

$$E(f) = \frac{b - a}{24} f''(c) h^2.$$

**Composite trapezoidal rule** Similarly, the *composite trapezoidal rule* is defined by

$$\int_a^b f(x) dx \approx \sum_{j=1}^n \frac{h_j}{2} [f(x_{j-1}) + f(x_j)].$$

For a uniform step-size  $h_j = h$ , this rule may be written as

$$\int_a^b f(x) dx \approx h \left( \frac{1}{2} f(x_0) + f(x_1) + f(x_2) + \dots + f(x_{n-1}) + \frac{1}{2} f(x_n) \right),$$

and the error has the representation

$$E(f) = -\frac{b-a}{12} f''(c) h^2.$$

**Composite Simpson rule** Likewise, the *composite Simpson rule* is

$$\int_a^b f(x) dx \approx \sum_{j=1}^n \frac{h_j}{6} [f(x_{j-1}) + 4f(x_{j-1/2}) + f(x_j)].$$

In particular, for a uniform step-size,

$$\begin{aligned} \int_a^b f(x) dx \approx \frac{h}{6} & \left( f(x_0) + 4f(x_{1/2}) + 2f(x_1) + 4f(x_{3/2}) + 2f(x_2) + \cdots \right. \\ & \left. + 2f(x_{n-1}) + 4f(x_{n-1/2}) + f(x_n) \right), \end{aligned}$$

and the error has the representation

$$E(f) = -\frac{b-a}{2880} f^{(4)}(c) h^4.$$

In view of the error representations derived, we expect that *for smooth integrands* Simpson's rule will be much more accurate than the midpoint or trapezoidal rules. To apply the composite Simpson's rule, it is best to compute separately

$$M_h(f) = \sum_{j=1}^n h_j f(x_{j-1/2}) \quad \text{and} \quad T_h(f) = \sum_{j=1}^n \frac{h_j}{2} [f(x_{j-1}) + f(x_j)],$$

and then combine these answers to obtain

$$S_h(f) = \frac{2}{3}M_h(f) + \frac{1}{3}T_h(f).$$

#### MATLAB script

```
function [S, M, T] = simpson(f, x)
% [S, M, T] = simpson(f, x) applies the Simpson, midpoint
% and trapezoidal rules to f(x) using the partition x.
h = diff(x);
xhalf = x(1:end-1) + h/2;
M = sum(h .* f(xhalf));
T = sum((h/2) .* ( f(x(1:end-1)) + f(x(2:end)) ) );
S = ( 2 * M + T ) / 3;
```

A simple use of the above script is listed below.

### MATLAB script

```
>> f = @(x) 1./(1+x.^2)
>> x = linspace(0,sqrt(3),100);
>> [S,M,T] = simpson(f,x)
>>
```

An equivalent Python script is listed below.

### Python script

```
import numpy as np
def simpson(f,x):
    h = np.diff(x)
    xhalf = x[0:-1] + h/2
    M = np.sum(h * f(xhalf))
    T = np.sum( (h/2)* (f(x[0:-1]) + f(x[1:])) )
    S = (2*M+T)/3
    return S,M,T

# main program
f = lambda x: 1/(1+x**2)
x = np.linspace(0,np.sqrt(3),100)
S,M,T = simpson(f,x)
print('S= %.10f M= %.10f T=%.16f' % (S,M,T))
```

The table below shows results for the integral

$$\int_0^{\sqrt{3}} \frac{dx}{1+x^2} = \frac{\pi}{3} = 1.047197551196598. \quad (6.2.2)$$

| $n$ | $M_h(f)$   | $T_h(f)$   | $S_h(f)$          |
|-----|------------|------------|-------------------|
| 2   | 1.05152656 | 1.03613754 | 1.046396886400760 |
| 4   | 1.04887358 | 1.04383205 | 1.047193069390340 |
| 6   | 1.04794657 | 1.04569717 | 1.047196767960434 |
| 8   | 1.04761955 | 1.04635281 | 1.047197303446330 |
| 10  | 1.04746783 | 1.04665669 | 1.047197449712940 |
| 20  | 1.04726519 | 1.04706226 | 1.047197544853653 |
| 40  | 1.04721446 | 1.04716372 | 1.047197550800163 |
| 80  | 1.04720178 | 1.04718909 | 1.047197551171821 |
| 160 | 1.04719861 | 1.04719544 | 1.047197551195049 |

The convergence behaviour is clearer if we look at the errors in  $M_h(f)$ ,  $T_h(f)$  and  $S_h(f)$ :

| $n$ | Midpoint   | Trapezoidal | Simpson            |
|-----|------------|-------------|--------------------|
| 2   | 0.00432901 | -0.01106001 | -0.000800664795838 |
| 4   | 0.00167603 | -0.00336550 | -0.000004481806257 |
| 6   | 0.00074902 | -0.00150038 | -0.000000783236163 |
| 8   | 0.00042200 | -0.00084474 | -0.000000247750267 |
| 10  | 0.00027028 | -0.00054086 | -0.000000101483657 |
| 20  | 0.00006764 | -0.00013529 | -0.000000006342945 |
| 40  | 0.00001691 | -0.00003383 | -0.000000000396435 |
| 80  | 0.00000423 | -0.00000846 | -0.000000000024777 |
| 160 | 0.00000106 | -0.00000211 | -0.000000000001548 |

In general  $E_h$ , if the error for some numerical method behaves like

$$E_h \approx Ch^\rho$$

for some  $\rho > 0$ , then

$$\rho \approx \frac{\log(E_{h_1}/E_{h_2})}{\log(h_1/h_2)}.$$

By computing the expression on the right, we can confirm experimentally the value of the exponent  $\rho$ . The table shows the errors and the observed exponent  $\rho$  such that  $E_h(f) \approx Ch^\rho$  for the integral (6.2.2).

| $n$ | Midpoint |        | Trapezoidal |        | Simpson   |        |
|-----|----------|--------|-------------|--------|-----------|--------|
| 4   | 1.68e-03 | (1.37) | -3.37e-03   | (1.72) | -4.48e-06 | (7.48) |
| 6   | 7.49e-04 | (1.99) | -1.50e-03   | (1.99) | -7.83e-07 | (4.30) |
| 8   | 4.22e-04 | (1.99) | -8.45e-04   | (2.00) | -2.48e-07 | (4.00) |
| 10  | 2.70e-04 | (2.00) | -5.41e-04   | (2.00) | -1.01e-07 | (4.00) |
| 20  | 6.76e-05 | (2.00) | -1.35e-04   | (2.00) | -6.34e-09 | (4.00) |
| 40  | 1.69e-05 | (2.00) | -3.38e-05   | (2.00) | -3.96e-10 | (4.00) |
| 80  | 4.23e-06 | (2.00) | -8.46e-06   | (2.00) | -2.48e-11 | (4.00) |
| 160 | 1.06e-06 | (2.00) | -2.11e-06   | (2.00) | -1.55e-12 | (4.00) |

## 6.3 Gaussian quadrature

In a Newton-Cotes rule we choose the points  $x_1, \dots, x_m$  *a priori* and find the weights  $w_1, \dots, w_m$  such that the quadrature is **exact** for polynomials of degree  $m$ , i.e., the degree of precision of the rule is at least  $m$ .

In a Gaussian quadrature, we choose points and weights such that the degree of precision is higher, e.g.,  $2m - 1$ .

Gaussian quadratures are established for integrals over  $[-1, 1]$ . Rules for integrals over  $[a, b]$  can be obtained via an affine mapping that maps  $[a, b]$  to  $[-1, 1]$ .

**Example 6.3.1** For  $m = 2$  a Gaussian rule for  $\int_{-1}^1 f(x) dx$  that has degree of precision at least  $2m - 1 = 3$  satisfies

$$\begin{aligned} \int_{-1}^1 dx &= w_1 + w_2 \\ \int_{-1}^1 x dx &= w_1 x_1 + w_2 x_2 \\ \int_{-1}^1 x^2 dx &= w_1 x_1^2 + w_2 x_2^2 \\ \int_{-1}^1 x^3 dx &= w_1 x_1^3 + w_2 x_2^3. \end{aligned}$$

The solution of this system of nonlinear equations is

$$w_1 = w_2 = 1, \quad x_1 = -\frac{1}{\sqrt{3}}, \quad x_2 = \frac{1}{\sqrt{3}}.$$

**Theorem 6.3.1 (Gauss-Legendre two-point rule)** *If  $f$  is continuous on  $[-1, 1]$  then*

$$\int_{-1}^1 f(x) dx \approx G_2(f) := f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right).$$

*This rule has degree of precision 3. Moreover, if  $f \in C^4[-1, 1]$ , then*

$$I(f) - G_2(f) = \frac{f^{(4)}(c)}{135}, \quad \text{for some } c \in (-1, 1).$$

**Example 6.3.2** Use the two-point Gauss-Legendre rule to approximate

$$\int_{-1}^1 \frac{dx}{x+2}.$$

Compare the result with those given by the composite trapezoidal rule with  $h = 1/2$  and Simpson's rule with  $h = 1$ .

Exact value

$$\int_{-1}^1 \frac{dx}{x+2} = \log(3) = 1.098612288668110.$$

Using two-point Gauss-Legendre rule:

$$\begin{aligned} \int_{-1}^1 \frac{dx}{x+2} &\approx f(-1/\sqrt{3}) + f(1/\sqrt{3}) \\ &= \frac{1}{-1/\sqrt{3}+2} + \frac{1}{1/\sqrt{3}+2} = \frac{12}{11} = 1.090909090909091 \end{aligned}$$

Using composite trapezoidal rule with  $h = 1/2$ :

$$\begin{aligned} \int_{-1}^1 \frac{dx}{x+2} &\approx h \left( \frac{1}{2}f(-1) + f(-0.5) + f(0) + f(0.5) + \frac{1}{2}f(1) \right) \\ &= \frac{67}{60} = 1.116666666666667 \end{aligned}$$

Using Simpson's rule with  $h = 1$ :

$$\begin{aligned} \int_{-1}^1 \frac{dx}{x+2} &\approx \frac{h}{6}(f(-1) + 4f(-0.5) + 2f(0) + 4f(0.5) + f(1)) \\ &= \frac{11}{10} = 1.10 \end{aligned}$$

**$N$ -point Gauss-Legendre rule** The  $N$ -point Gauss-Legendre rule for  $\int_{-1}^1 f(x) dx$  is given by

$$G_N(f) = \sum_{i=1}^N w_{N,i} f(x_{N,i}).$$

The weights  $w_{N,i}$  and **abscissas**  $x_{N,i}$ ,  $i = 1, \dots, N$  are given in a table.

The error is

$$I(f) - G_N(f) = e_N \frac{f^{(2N)}(\xi)}{(2N)!}$$

where  $\xi \in (-1, 1)$  and

$$e_N = \frac{2^{2N+1}(N!)^4}{(2N+1)[(2N)!]^2} \approx \frac{\pi}{4^N} \quad \text{as } N \rightarrow \infty.$$

### Gauss-Legendre abscissas and weights

The previous table in rational numbers is given below.

| $N$ | $x_{N,i}$               | $w_{N,i}$         |
|-----|-------------------------|-------------------|
| 2   | $\pm 0.577350269189626$ | 1.000000000000000 |
| 3   | $\pm 0.774596669241483$ | 0.555555555555556 |
|     | 0.000000000000000       | 0.888888888888889 |
| 4   | $\pm 0.861136311594053$ | 0.347854845137454 |
|     | $\pm 0.339981043584856$ | 0.652145154862546 |

| $N$ | $x_{N,i}$                        | $w_{N,i}$             |
|-----|----------------------------------|-----------------------|
| 2   | $\pm 1/\sqrt{3}$                 | 1                     |
| 3   | $\pm \sqrt{3/5}$                 | 5/9                   |
|     | 0                                | 8/9                   |
| 4   | $\pm \sqrt{(3 + 2\sqrt{6/5})/7}$ | $(18 - \sqrt{30})/36$ |
|     | $\pm \sqrt{(3 - 2\sqrt{6/5})/7}$ | $(18 + \sqrt{30})/36$ |

**Integrals over  $[a, b]$**  Use the change of variable

$$t = \frac{a+b}{2} + \frac{b-a}{2}x.$$

Then

$$\int_a^b f(t) dt = \int_{-1}^1 f\left(\frac{a+b}{2} + \frac{b-a}{2}x\right) \frac{b-a}{2} dx.$$

Hence

$$\int_a^b f(t) dt \approx \frac{b-a}{2} \sum_{i=1}^N w_{N,i} f\left(\frac{a+b}{2} + \frac{b-a}{2}x_{N,i}\right).$$

## 6.4 Difficult integrals

In the following example the trapezoidal rule and Simpson's rule do not have the expected orders of convergence. Why?

**Example 6.4.1** Approximate the integral

$$I = \int_0^1 \sqrt{x} dx = \frac{2}{3}$$

using the trapezoidal rule and Simpson's rule with  $N = 2, 4, 8, 16, 32, 64$ .

| $n$ | Trapezoidal rule |          |        | Simpson's rule |          |        |
|-----|------------------|----------|--------|----------------|----------|--------|
|     | $T_h(f)$         | $E_h(f)$ |        | $S_h(f)$       | $E_h(f)$ |        |
| 2   | 0.6035533906     | 6.31e-02 |        | 0.6380711875   | 2.86e-02 |        |
| 4   | 0.6432830462     | 2.34e-02 | (1.43) | 0.6565262648   | 1.01e-02 | (1.50) |
| 8   | 0.6581302216     | 8.54e-03 | (1.45) | 0.6630792801   | 3.59e-03 | (1.49) |
| 16  | 0.6635811969     | 3.09e-03 | (1.47) | 0.6653981886   | 1.27e-03 | (1.50) |
| 32  | 0.6655589363     | 1.11e-03 | (1.48) | 0.6662181827   | 4.48e-04 | (1.50) |
| 64  | 0.6662708114     | 3.96e-04 | (1.49) | 0.6665081031   | 1.59e-04 | (1.49) |

In general, the function

$$f(x) = \sqrt{x} \text{ on } [0, b]$$

has unbounded derivatives on  $[a, b]$ .

To get around the problem, we introduce the change of variables:  $x = y^2$  so  $dx = 2ydy$  and

$$\int_0^b \sqrt{x} dx = 2 \int_0^{\sqrt{b}} y^2 dy$$

**Example 6.4.2**

$$f(x) = \sqrt{|x - 1/4|} \text{ on } [0, 1]$$

First, we split integral to remove the discontinuity of the derivative from  $|\cdot|$

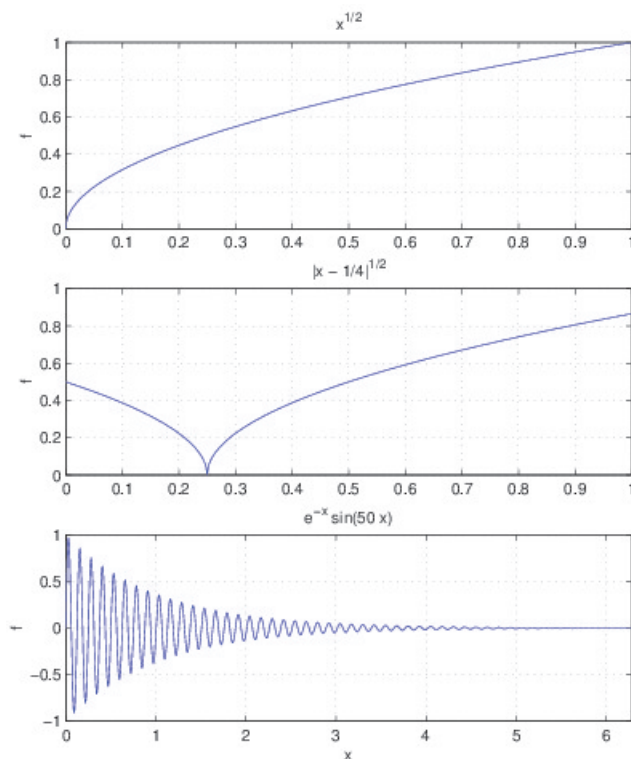
$$\int_0^1 f(x) dx = \int_0^{\frac{1}{4}} \sqrt{1/4 - x} dx + \int_{\frac{1}{4}}^1 \sqrt{x - 1/4} dx$$

Then use change of variables to remove square roots.

**Example 6.4.3** For this oscillatory integral,

$$f(x) = e^{-x} \sin(50x)$$

we have to use  $n$  sufficiently large.



**MATLAB function quad** The MATLAB function `quad` applies Simpson's rule to a given integral, choosing the points automatically. For instance, we can evaluate (6.2.2) as follows:

MATLAB script

```
>> f=@(x)1./(1+x.^2);  
>> quad(f,0,sqrt(3))  
  
ans = 1.04719749069872
```

By default, `quad` uses an absolute error tolerance of  $10^{-6}$ , but you can change this with an optional fourth argument.

MATLAB commands

```
>> quad(f,0,sqrt(3),1e-12)  
  
ans = 1.04719755119660
```

In Python, the function `quad` in the `scipy.integrate` package performs a similar task.

Python script

```
from scipy import integrate  
import numpy as np  
  
f= lambda x:1/(1+x**2)  
res = integrate.quad(f, 0, np.sqrt(3))  
print('Result = %2.10f ' % res[0])
```